

FALCONLEDGER

PoPL Lending Protocol Blueprint

Proof of Participation & Liquidity

Native Fixed-Rate Open-Ended Lending with Protocol Emissions

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Status	Design Phase - Ready for Implementation Planning
Related	Falcon-faucet-wallet, AMM Pools, Quantum-Resistant Ledger

1. Executive Summary

This blueprint defines a **native protocol-level lending system** for FalconLedger, a quantum-resistant fork of the XRP Ledger. The system introduces open-ended fixed-rate lending pools with over-collateralization, combined with a powerful **Proof of Participation & Liquidity (PoPL)** incentive mechanism.

Key Innovations:

- **Open-ended fixed-rate lending** — Borrowers get predictable rates without hard maturity dates; repay anytime.
- **Native over-collateralized model** — Falcon as primary collateral for borrowing FUsd (or other pool assets).
- **PoPL Emissions** — Fixed 0.5% network emission split between validators and liquidity providers. LPs earn Falcon proportionally to their share of supplied liquidity (no oracle dependency for rewards).
- **Dual yield for LPs** — Organic interest from borrowers + protocol Falcon emissions.
- **Reuses existing AMM infrastructure** — Share accounting, pool UI patterns, and liquidation paths from current AMM pools.

The design prioritizes simplicity, auditability, and strong bootstrap incentives for mainnet launch while maintaining the security and performance characteristics of the underlying ledger.

2. Vision: Proof of Participation & Liquidity (PoPL)

FalconLedger evolves beyond traditional validator-only rewards. In the PoPL model, economic security (validators) and economic activity (liquidity provision) are both first-class citizens rewarded from a fixed emission pool.

This creates a flywheel: More liquidity → more borrowing capacity → more organic interest + protocol emissions → higher effective APY → more liquidity.

Emission Philosophy

Total network emission remains fixed at 0.5%. The split between validators and LPs is a governance-configurable parameter. Early in the network's life, a larger share goes to LPs to bootstrap real usage. Over time, as organic interest volume grows, the LP allocation can be tapered.

3. System Overview

Core Components

- **LendingPool** — Protocol object representing a supply pool for a specific asset (e.g. FUsd). Tracks total supplied, borrowed, interest accrued, and LP shares.
- **LendingPosition / Shares** — Per-user share ownership in a pool. Used for both interest distribution and PoPL emissions.
- **Collateral Tracking** — Separate from the pool. Borrowers post Falcon (or supported assets) as collateral against borrowed FUsd.

- **Reward Accumulator** — Tracks claimable Falcon emissions per LP share. Updated periodically or on claim.
- **Health Factor Engine** — Calculates collateralization ratio in real time for liquidation triggers.

The system is designed so that the majority of logic lives in the C++ daemon (rippled fork), with the TypeScript wallet/client providing the user interface and transaction construction (reusing existing patterns from Falcon-faucet-wallet).

4. Core Lending Mechanics

4.1 Supply Side (Liquidity Providers)

Users supply the pool's primary asset (initially FUsd) and receive LendingShares proportional to their deposit. They earn two revenue streams: (1) interest paid by borrowers, and (2) Falcon emissions from the PoPL program.

4.2 Borrow Side

Borrowers post collateral (primarily Falcon) and borrow the pool asset (FUsd). The position is over-collateralized at a configurable ratio (target: 150% minimum). Interest accrues continuously based on ledger close time using the pool's fixed APR.

Example — 150% Over-Collateralization

Action	Details
Borrow	1,000 FUsd
Collateral Required (150%)	~1,500 FUsd worth of Falcon (at oracle price)
Fixed APR	5% (set at pool creation)
Accrual	Continuous — calculated on repay/claim using ledger time delta

4.3 Interest Accrual (Open-Ended)

No hard maturity. Interest is calculated on every relevant transaction (repay, partial repay, liquidation, claim) using:

$$\text{Accrued} = \text{Principal} \times (\text{APR} / \text{SecondsPerYear}) \times \Delta\text{LedgerCloseTime}$$

This keeps the system simple while giving borrowers flexibility.

5. PoPL Emissions & LP Rewards

5.1 Fixed Emission Pool

The network maintains a fixed emission rate (currently parameterized at 0.5%). This emission is split between:

- Validators (security / consensus)

- Liquidity Providers in incentivized pools (lending + optionally AMM)
- Other buckets (development, community treasury, reserves) — configurable

5.2 LP Reward Calculation (Share-Ratio Only)

Rewards are distributed purely based on ownership share within each pool — **no USD valuation or oracles required** for the emissions math. This reuses the same share accounting used for interest distribution.

Example (Total supplied = 1,000 units)

Provider	Supplied	Share	Emissions %
LP1	500	50%	50% of lending emissions
LP2	200	20%	20%
LP3	200	20%	20%
LP4	100	10%	10%

Formula: $\text{FalconReward_user} = E_lending \times (\text{UserShares} / \text{TotalShares})$

5.3 Dual Yield

LPs receive the sum of (a) their pro-rata share of borrower-paid interest and (b) their pro-rata Falcon emissions. This creates highly competitive effective APY, especially during the bootstrap phase.

6. Data Model (Ledger Objects)

6.1 LendingPool (new ledger entry type)

- PoolID / Key (derived from asset + parameters)
- SuppliedAsset (e.g. FUsd)
- TotalSupplied
- TotalBorrowed
- FixedAPR_bps
- LastAccrualTime
- TotalShares (for LP accounting)
- RewardAccumulator (for PoPL emissions)
- CollateralRequirements (min ratio, supported collateral assets)
- Status (active, paused, deprecated)

6.2 LendingPosition / Share Ownership

Implemented via account-owned objects or extended trustline-style entries. Tracks per-user share balance, last claim time for interest, and last claim time for emissions.

6.3 Collateral Positions

Separate tracking (not inside the LendingPool object) for clean risk isolation. Contains collateral asset/amount, associated borrow position, health factor snapshot, and maturity/liquidation state if applicable.

7. Transaction Types (New)

Tx Type	Purpose
LendingPoolCreate	Create a new supply pool with fixed rate and parameters
LendingSupply	Deposit asset, mint shares
LendingBorrow	Post collateral, draw funds (checks health factor)
LendingRepay	Repay principal + accrued interest (triggers accrual)
LendingWithdraw	Burn shares, withdraw available liquidity
LendingLiquidate	Liquidate under-collateralized positions
LendingClaimRewards	Claim accrued interest + Falcon emissions

8. Liquidation & Risk Management

Liquidation is triggered when a position's Health Factor falls below the configured threshold (typically 1.0–1.1). Anyone can initiate liquidation. The liquidator repays part or all of the debt and receives collateral at a discount (bonus). The repaid amount (including interest) flows back to the pool for LPs.

Because collateral (Falcon) and borrowed asset (FUsd) are different, liquidation can optionally route through the existing AMM for efficient swaps.

9. Tokenomics & Emissions Parameters

Final Parameters (Locked)

Parameter	Value	Notes
Emission Rate	0.5% per epoch	Recurring every epoch
Bootstrap Split (Epoch 1)	50% Validators / 50% LPs	Mainnet launch incentive
Taper Period	24 epochs (~6 months)	Linear reduction
LP Allocation Drop	20 percentage points total	50% → 30% LP share
Taper Step	≈ 0.833 pp reduction per epoch	Linear: $20 \div 24$
Final Steady State	70% Validators / 30% LPs	After epoch 24, permanent
Fixed Lending APR	5%	Pool-level fixed rate

10. Implementation Roadmap

- **Phase 0 (Current)** — Finalize blueprint, tokenomics parameters, and economic model. Community / validator alignment on PoPL split.
- **Phase 1** — Core lending objects + transactions in daemon (LendingPool, Supply/Borrow/Repay/Withdraw). Testnet deployment.
- **Phase 2** — Collateral & health factor engine + basic liquidation. PoPL reward accumulator (share-ratio only).
- **Phase 3** — Client integration in Falcon-faucet-wallet (new lending UI, borrow/repay flows, position dashboard). Full testnet E2E.
- **Phase 4** — Security review, invariant testing, economic simulations. Amendment proposal & validator voting.
- **Phase 5** — Mainnet activation with aggressive LP emissions for launch. Ongoing monitoring + taper schedule activation.

11. Security Considerations

- All new logic must pass through standard preclaim/apply/invariant checks.
- Health factor and liquidation math must be deterministic and resistant to manipulation.
- Reward distribution must be fair and resistant to flash-liquidity gaming (time-weighting or minimum participation periods can be added).
- Oracle usage is isolated to collateral pricing only — reward math does not depend on oracles.
- Full security report and audit process (following the pattern already used for AMM/bridge features).

This blueprint represents the current consensus design. It is intended as a living document that will be refined during implementation and community review.